

WILL BORDEN

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EDUCATION

The University of Melbourne · MEng Electrical Engineering Mar 2026 – Expected Dec 2028
Weighted Average Mark: 87.9

The University of Melbourne · BSci Electrical Engineering Systems Feb 2022 – Dec 2025
Weighted Average Mark: 87.3 · Dean's Honours List (2022) · Diploma in Languages – French · Academic Exchange, The University of Bordeaux, Aug 2024 – Dec 2024

EXPERIENCE

Melbourne Space Laboratory Jul 2025 – Present
Research Assistant - Electronics Engineering Melbourne

- Designed thermal control unit for ASA lunar rover, carrying a NASA payload to the southern lunar pole.
- Implemented an NVIDIA Jetson-based satellite camera system, launching on a Gilmour Space bus in June 2026.
- Designed and manufactured complex, EMI compliant PCBs for use with radiation-hardened electronic components, including detailed failure mode analysis and error budget calculations for lunar and orbital environments.
- Developed software for Linux and embedded devices, including RTOS-based firmware, designed unit tester for EM, QM, and FM hardware validation.

Esper Satellite Imagery Feb 2025 – Jul 2025
Electronics Engineer Melbourne

- Designed and validated space-grade on-board computer and power distribution system for hyperspectral Earth observation payload launching August 2026.
- Managed interface control documents and design reviews with external space providers to ensure compliance with mission requirements.
- Designed and outsourced space-grade wiring harnesses and PCBs; conducted extensive electrical failure mode analysis.

ARES Rocketry Team Feb 2022 – Present
Chief Engineer – 2023 - 2024, Lead Systems Engineer – 2025 - Present The University of Melbourne

- Led a multidisciplinary team of 100+ students to design and launch a 30,000 ft sounding rocket, achieving 4th place out of 152 universities at the 2024 Spaceport America Cup.
- Designed, implemented, and tested an avionics flight computer for parachute deployment and in-flight data logging; selected for podium presentation at the 2023 Spaceport America Cup (20 of 120 submissions).
- Currently serving as Lead Systems Engineer for Project Odyssey, ARES' first entirely student-designed hybrid rocket, applying industry-standard aerospace systems engineering practices.

QinetiQ Australia Feb 2023 – Present
Technical Staff – University of Melbourne Creator Space Melbourne

- Supervised and mentored students and professional researchers in a state-of-the-art makerspace, providing expert training in electronics, PCB debugging, soldering, and test equipment operation.
- Managed technical competency tracking programs and supported safety protocols for workshop operations and events.
- Developed and ran bottle rocket outreach program for women and Indigenous students in STEM programs.

TECHNICAL SKILLS

PCB & Circuit Design: Multi-layer PCB design in Altium Designer (7 yrs), KiCAD, EagleCAD; analog, high-speed, RF & embedded systems; radiation-tolerant design for LEO/lunar; EMI/EMC to MIL-STD-461; SPICE/TINA-TI/QUCS simulation

Embedded Systems: Embedded C/C++ (5 yrs); RTOS frameworks; driver development; PID & Kalman control; FPGA (Verilog/Quartus); protocols: I2C, SPI, UART, USB, CAN, Ethernet, PCIe, MIPI CSI-2; Git, Linux

Systems Engineering: Failure mode & reliability analysis; test campaign design & hardware validation; requirements management; interface control documentation; project leadership (100+ member teams)

Tools & Equipment: Oscilloscopes, function generators, spectrum analyzers, VNAs; Atlassian project management suite; Advanced SMT assembly & hand soldering

AWARDS & HONORS

4th Place (of 152 teams) – Spaceport America Cup 2024 · Podium Presentation Selection – Spaceport America Cup 2023 (20 of 120) · Dean's Honours List, University of Melbourne 2022 · Melbourne Global Scholar's Award 2024 · Alan Gilbert Mobility Scholarship 2024